

• By condition 3 of the ~~pref~~ pref, for each column having a pivot, all positions below it must contain 0. By condition 5 of pref, for each column having a pivot, all positions above it must contain 0.

$\therefore I$ is the pref of A .

By Theorem 7, A is invertible. $\square$

Exercises 25 and 26 prove Theorem 4 for $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$.

25) Show that if $ad - bc = 0$, then the equation $Ax = 0$ has more than one solution. Why does this imply that A is not invertible?

Solution: By Theorem 3 from Chapter 1, the matrix equation $Ax = 0$ has the same solution set as the system of equations corresponding to the augmented matrix $[A \ 0]$.

Let $a \neq 0$.
$$\begin{bmatrix} a & b & 0 \\ c & d & 0 \end{bmatrix} \xrightarrow[\times 1/a]{\text{Row 1} = \text{Row 1}} \begin{bmatrix} 1 & b/a & 0 \\ c & d & 0 \end{bmatrix} \xrightarrow[-c \times \text{Row 1}]{\text{Row 2} = \text{Row 2}} \begin{bmatrix} 1 & b/a & 0 \\ 0 & d - \frac{cb}{a} & 0 \end{bmatrix}$$

It's very easy to verify that the final matrix is in echelon form.

If $d - \frac{cb}{a} = 0$, then Column 2 of A has no pivots. By Theorem 2 from Chapter 1, the system of equations has infinite solutions.

$d - \frac{cb}{a} = 0 \Rightarrow \boxed{ad - cb = 0}$.

$\therefore A$ cannot be reduced to I when $a \neq 0$ and $ad - cb = 0$. By Theorem 7, A is not invertible.

Let $a = 0$. $\therefore$ since $ad - bc = 0$, at least one of b or c must be 0.

Let $b = 0$. $\therefore$ $\begin{bmatrix} a & b & 0 \\ c & d & 0 \end{bmatrix} \xrightarrow[\text{Row 2}]{\text{Interchange Row 1 and Row 2}} \begin{bmatrix} c & d & 0 \\ 0 & 0 & 0 \end{bmatrix}$. By similar reasoning as above, A is not invertible.

Let $c = 0$. $\begin{bmatrix} 0 & b & 0 \\ 0 & d & 0 \end{bmatrix}$. Column 1 has no pivots. $\therefore A$ cannot be reduced to I . By Theorem 7, A is not invertible. $\square$